

SOFTWARE REQUIREMENTS SPECIFICATION

Attendance Tracker – Single-Photo Classroom Attendance

Modified, Updated and Consolidated SRS — Version 2.0

Document Control

Item	Value
Product	Attendance Tracker
Status	Updated / Authoritative
Platforms	Web, Android, iOS
Primary roles	Admin, Faculty, Student
Attendance input	Single classroom photograph
Recognition scope	Students enrolled in one or more faculty-selected classes
Final authority	Faculty

1. Introduction

1.1 Purpose

Attendance Tracker is a classroom attendance management application that reduces manual attendance work using a single classroom photograph. Faculty select one or more assigned classes, capture a photograph, and receive an automatically generated attendance result based only on students eligible through those selected classes. Automated recognition assists the process; faculty remain the final authority and can review, correct, finalize and later edit attendance.

1.2 Scope

- Authentication and role-based authorization.
- Student, faculty and class management.
- Faculty-to-class assignment and student enrollment.
- Student profiles and face-enrollment status.
- Selection of one or multiple assigned classes for an attendance session.
- Single-photo classroom attendance capture.
- Backend face detection/recognition as the eventual biometric capability.
- Class-scoped recognition candidate selection.
- Low-confidence, unknown and twin/ambiguous identity review.
- Faculty correction, finalization and post-finalization editing.
- Attendance history, reports, Admin oversight, audit and institution settings.
- Pagination/incremental loading for large collections.

1.3 Core Principles

- Faculty selects one or more assigned classes.
- Single-class attendance remains supported as the one-class case of the same workflow.
- The candidate pool is the union of students enrolled in the selected classes only.
- Students belonging only to unselected classes must never become eligible candidates.
- The frontend must not load the entire institution student database for recognition.
- AI status/confidence and final faculty-confirmed attendance are separate.

- Faculty is the final attendance authority.
- Ambiguous/twin cases remain REVIEW until resolved.
- Finalized attendance remains editable by appropriately authorized faculty, with the required reason/authorization.
- Manual changes are auditable and cannot create duplicate (session_id, student_id) records.

2. Product Overview

2.1 Roles

Role	Responsibilities	Scope
Admin	Institution-level management, oversight, faculty/student/class management, reports, audit and settings.	Institution-wide.
Faculty	Assigned-class operations, attendance capture/review/edit/finalization.	Assigned classes and authorized students only.
Student	Student entity/data representation.	No privileged administrative capability in current scope.

2.2 Platforms

- Support Web, Android and iOS.
- Use responsive/adaptive layouts for different screen sizes and input methods.
- Enforce authorization at the backend/service layer, not only through UI visibility.

3. Architecture and Face-Enrollment Boundary

The current frontend and eventual biometric backend are separate concerns.

3.1 Current Frontend

- Provide face-enrollment UI/status.
- Do not perform biometric processing.
- Do not generate embeddings.
- Do not inspect biometric vectors.
- Do not store biometric templates in the frontend.
- Do not implement face recognition in the frontend.

3.2 Eventual Recognition Backend

- Perform face detection and recognition.
- Use the selected and validated face-recognition architecture/model; a dedicated pretrained face-recognition model such as an ArcFace/InsightFace-family model may be used subject to final validation and commercial-use requirements.
- Use OpenCV for applicable image operations.
- Use class-scoped candidate matching.
- Retain model/version information where supported.
- Calibrate recognition thresholds and ambiguity margins using representative classroom data.

4. Functional Requirements

ID	Area	Requirement
FR-01	Authentication	Users authenticate before protected functions.
FR-02	Authorization	Admin, Faculty and Student permissions are enforced by backend/service controls.
FR-03	Student List	Search/filter by applicable class, department, semester and low-attendance criteria; support pagination/incremental loading.
FR-04	Student Profile	Show profile image, name, student ID, roll number, department, semester, section, enrolled classes, overall attendance, attendance by class, recent attendance and face-enrollment status.
FR-05	Student Enrollment	Authorized users manage student enrollment in classes.
FR-06	Faculty Management	Admin can list, search/filter, create, edit and view faculty and maintain active/inactive/on-leave status.
FR-07	Faculty-Class Assignment	Admin assigns faculty to classes; faculty access is limited to authorized assignments.
FR-08	Class Management	Admin can list/create/edit/view classes with department, semester, section, academic session and applicable status.
FR-09	Class Roster	Authorized users can view enrolled students.
FR-10	Class Selection	Faculty can select one or multiple assigned classes for one attendance session.
FR-11	Candidate Scope	Recognition candidates are the union of students enrolled in the selected classes only.
FR-12	Photo Capture	Faculty captures one classroom photograph for the session.
FR-13	Recognition	Backend detects visible faces and compares them with eligible enrolled face data.
FR-14	Confidence	Reliable matches may prepopulate results; uncertain/unknown/ambiguous cases go to review.
FR-15	Twin Handling	System must not choose between highly similar/twin candidates merely because one score is slightly higher.
FR-16	Twin Actions	Applicable cases support Both Present, Student A only, Student B only, or Decide Later.

FR-17	States	PRESENT, ABSENT, UNKNOWN and REVIEW remain supported.
FR-18	AI/Final Separation	AI status/confidence and final faculty-confirmed status are stored separately.
FR-19	Editing	Faculty can edit attendance before finalization.
FR-20	Post-Finalization	Authorized faculty can edit finalized attendance; appropriate reason/authorization is required.
FR-21	Duplicate Prevention	Editing does not create duplicate (session_id, student_id) records.
FR-22	History	Authorized users can view historical attendance with appropriate filtering.
FR-23	Reports	Faculty and Admin reports follow Section 10.
FR-24	Admin Oversight	Admin can inspect institution-wide attendance without bypassing review/finalization rules.
FR-25	Audit	Relevant administrative and attendance changes are auditable; audit viewing is read-only.
FR-26	Settings	Admin can manage supported institution settings, including attendance threshold.

5. Attendance Workflow

1. Faculty authenticates.
2. The system shows classes assigned to that faculty member.
3. Faculty selects one or more assigned classes.
4. The attendance session records its selected class scope.
5. Eligible candidates are the union of enrolled students in the selected classes.
6. Faculty captures one classroom photograph.
7. The recognition backend detects faces and evaluates only eligible candidates.
8. Reliable matches generate preliminary attendance.
9. Low-confidence, UNKNOWN or ambiguous results are marked REVIEW rather than silently finalized.
10. Twin/near-identical cases present the defined faculty review actions.
11. Results can be grouped/filtered by class where applicable.
12. Faculty reviews and edits attendance.
13. Faculty finalizes/submits the session.
14. Authorized faculty may later edit finalized attendance with the appropriate reason/authorization.
15. The original AI result remains preserved and the latest final result represents the authorized faculty decision.

6. Attendance States and Rules

State	Meaning	Rule
PRESENT	Confirmed presence.	Can appear as AI or final status.

ABSENT	No confirmed presence.	May be a final faculty decision; unresolved recognition must not be treated as absence automatically.
UNKNOWN	No sufficiently reliable identity.	Must not be silently converted to PRESENT/ABSENT.
REVIEW	System cannot safely make final decision due to ambiguity, low confidence or twin handling.	Remains REVIEW until faculty resolves it; AI result remains unchanged.

AI status is the original automated result. Final status is the faculty-confirmed result. Neither REVIEW nor UNKNOWN is an ordinary final outcome until faculty explicitly resolves the case.

7. Twin / Ambiguous Identity

Twin/near-identical candidates may belong to different selected classes in a multi-class session. The system must treat ambiguity explicitly.

16. Identify strongest candidate identities.
17. Evaluate configured ambiguity margin.
18. Set ambiguous result to REVIEW.
19. Preserve original AI result/confidence.
20. Offer Both Present, Student A only, Student B only or Decide Later, as applicable.
21. Keep REVIEW when Decide Later is selected.
22. Store the faculty decision separately as final status.

A student belonging only to an unselected class is never an eligible candidate, even if visually similar.

8. Student, Faculty and Class Management

8.1 Student

- Student list: search, class/department/semester/low-attendance filters, pagination/incremental loading.
- Profile: profile image, name, student ID, roll number, department, semester, section, enrolled classes, overall attendance, attendance by class, recent attendance, face-enrollment status.

8.2 Faculty

- List, search/filter, profile, create, edit, status and faculty-to-class assignment.
- Statuses include active, inactive and on-leave.
- Faculty access is limited to authorized classes/students.

8.3 Class

- List, create, edit, detail, department, semester, section, academic session, faculty assignment, student enrollment, roster and applicable archived/status handling.

9. Attendance History

- View previous attendance sessions.
- Class filtering.
- Student filtering where applicable.
- Faculty filtering for authorized administrative users.

- Status filtering.
- Date filtering.
- Search where applicable.
- Faculty sees only authorized scope; Admin has institution-level oversight.
- Large collections use pagination/incremental loading.

10. Reports

10.1 Faculty Reports

- Overall attendance percentage.
- Total attendance sessions.
- Attendance trend over time.
- Attendance by class.
- Per-student attendance percentage.
- Attended/total session counts.
- Low-attendance students.
- Institution-configured attendance threshold.
- Class-scoped reporting.

10.2 Admin Reports

- Institution-wide attendance.
- Department/class/faculty breakdown where applicable.
- Per-student statistics.
- Low-attendance students.
- Attendance trends.

10.3 Reporting Rules

- Threshold comes from institution settings and is not hard-coded in UI.
- REVIEW and UNKNOWN are not treated as ordinary attendance outcomes.
- Final faculty-confirmed results are authoritative for statistics once resolved.
- Report student statistics support pagination/incremental loading.

DEFERRED/PLANNED: the final report date-range picker UI is not finalized. Report/data export is not currently implemented and must not be presented as available functionality.

11. Admin Application

- Admin Dashboard.
- Faculty Management: list, create, edit, profile, status, assignment.
- Student Management: list, profile, enrollment information.
- Class Management: list, create, edit, detail, faculty assignment and enrollment.
- Attendance: institution-wide oversight, session inspection and attendance-record inspection.
- Reports: institution-level analytics.
- Audit: read-only audit history.
- Settings: institution settings, attendance threshold and other supported configuration.

Admin must not bypass the established attendance review/finalization workflow simply because of the Admin role.

12. Authentication and Authorization

- Admin: institution-level management and oversight.
- Faculty: assigned-class operations only; attendance capture/review/editing for authorized classes.
- Student: no privileged administrative capability in the current scope.
- Backend/service authorization is mandatory.
- Faculty access to unrelated classes, students or attendance must be rejected by the service/backend.

13. Audit

- Faculty created.
- Faculty updated.
- Class created.
- Class updated.
- Faculty assigned to class.
- Institution settings changed.
- Attendance status changed.
- Post-finalization attendance edited.
- Other relevant administrative actions.

13.1 Audit Entry

- Actor.
- Timestamp.
- Action.
- Entity type.
- Entity ID.
- Entity label.
- Previous value where applicable.
- New value where applicable.
- Relevant attendance/session/student information where applicable.

Audit history is read-only for users viewing it.

14. Pagination and Scalability

- Students, faculty, classes, attendance history, audit entries and report student statistics support pagination/incremental loading.
- Filtering occurs before pagination.
- Pagination must not create duplicates, omissions or incorrect counts.
- Recognition requests must not require loading the institution-wide student database into the frontend.

15. Database Requirements

The existing normalized database remains the foundation. No unrelated business entities are introduced and the database is not replaced.

Existing Entity/Table	Purpose
Users	Authentication identity and role.
Faculty	Faculty profile and class authorization.
Students	Student master/academic information.

Classes	Class/subject, department, semester, section and academic session.
Faculty-Class assignments	Defines faculty authorization for classes.
Enrollments	Defines students belonging to classes.
Attendance Sessions	Represents an attendance capture event and selected class scope.
Attendance Records	Stores per-student attendance within a session.
Student face-enrollment data	Represents face-enrollment data/status; biometric processing belongs to backend recognition.

15.1 Multi-Class Session

- A session may contain one or multiple selected classes.
- Selected class IDs are represented by the finalized attendance-session model.
- Candidate pool is the union of students enrolled in selected classes.
- Unselected-class-only students cannot receive attendance from the session.
- Single-class attendance remains the one-class case.
- Multi-class attendance extends the existing attendance/results architecture.
- Results are grouped/filterable by class where applicable.

15.2 Integrity

- Primary/foreign keys enforce existing relationships.
- Assignments and enrollments prevent invalid/duplicate relationships.
- (session_id, student_id) is unique for attendance records.
- Session class scope is validated against faculty authorization.
- AI and final attendance information remain logically distinct.
- Status/confidence constraints are enforced by the finalized database implementation.

16. Attendance Editing and Finalization

16.1 Before Finalization

- Faculty can edit recognition-generated attendance.
- Faculty can resolve REVIEW cases.
- Faculty can correct PRESENT/ABSENT results.
- Original AI result remains preserved.

16.2 After Finalization

- Authorized faculty can edit finalized attendance.
- Appropriate reason/authorization is captured.
- Latest final result is preserved.
- Original AI result is unchanged.
- Modification is auditable.
- Existing record is updated rather than duplicated.

16.3 Final Authority

AI recognition is advisory. Faculty remains the final authority. Admin oversight does not remove or bypass the established review/finalization workflow.

17. Use Cases

ID	Use Case	Actor
UC-01	Authenticate User	Admin / Faculty / Student
UC-02	Create/Manage Faculty	Admin
UC-03	Assign Faculty to Class	Admin
UC-04	Manage Student	Admin
UC-05	View Student Profile	Admin / Faculty
UC-06	Enroll Student	Admin
UC-07	Manage Class	Admin
UC-08	Select One or Multiple Attendance Classes	Faculty
UC-09	Take Attendance Photo	Faculty
UC-10	Process Recognition	System / Backend
UC-11	Review Recognition	Faculty
UC-12	Resolve Twin/Ambiguous Identity	Faculty
UC-13	Edit Attendance	Faculty
UC-14	Finalize Attendance	Faculty
UC-15	Edit Finalized Attendance	Authorized Faculty
UC-16	View Attendance History	Admin / Faculty
UC-17	View Reports	Admin / Faculty
UC-18	Admin Attendance Oversight	Admin
UC-19	View Audit History	Authorized Viewer
UC-20	Manage Institution Settings	Admin

18. Non-Functional Requirements

Category	Requirement
Performance	Attendance processing shall fit a practical classroom workflow; actual targets are established by representative benchmarking.
Accuracy	Recognition thresholds shall be validated; false-positive identity assignment is a critical error.
Security	Authentication, authorization and protected storage are required.
Privacy	Face images/biometric-related data shall follow applicable institutional policy and law.
Reliability	Failure to recognize one face must not lose the whole attendance session.
Scalability	Use indexed access, efficient queries and pagination/incremental loading.
Maintainability	AI model/version and AI/final decision states shall be explicit.
Auditability	Administrative and attendance modifications shall be traceable.
Usability	Web, Android and iOS interfaces shall be responsive/adaptive.

19. Assumptions and Constraints

- Recognition quality depends on lighting, face visibility, pose, occlusion and enrollment quality.
- A single photograph may not expose every student's face clearly; faculty correction is therefore supported.

- Recognition thresholds require project-data calibration.
- Final embedding/vector dimensions must match the selected recognition model in the backend implementation.
- Accurate class enrollment data is required for correct candidate scoping.
- No additional business entity is required solely for twin handling or post-submission editing.

20. Deferred / Planned / Outside Current Scope

- Final report date-range picker UI — DEFERRED/PLANNED.
- Report/data export — DEFERRED/PLANNED.
- Frontend biometric processing — OUT OF CURRENT FRONTEND SCOPE.
- General-purpose LLM as the primary face-identity engine — OUT OF SCOPE.
- Unrelated business modules or a replacement database architecture — OUT OF SCOPE.

21. Acceptance Criteria

ID	Scenario	Expected Result
AC-01	Select CSE-5A → capture	Only CSE-5A students are eligible.
AC-02	Select CSE-5A + CSE-5B → capture	Only students from those two classes are eligible.
AC-03	Student belongs only to CSE-3A; CSE-3A not selected	Student must not receive attendance.
AC-04	Select one class	Works as the one-class form of the same session architecture.
AC-05	Faculty attempts unrelated class	Access is rejected.
AC-06	Two selected-class candidates are ambiguous	REVIEW; faculty gets Both Present / Student A only / Student B only / Decide Later.
AC-07	Decide Later	REVIEW remains and AI result is preserved.
AC-08	Faculty resolves review	Final status is stored without destroying AI result.
AC-09	Edit before finalization	Faculty correction updates final attendance.
AC-10	Finalize → later edit	Reason/authorization recorded; original AI result retained.
AC-11	Edit attendance	No duplicate (session_id, student_id) record is created.
AC-12	Faculty access	Faculty cannot access students/classes/attendance outside assigned scope.
AC-13	Admin oversight	Admin can inspect institution-wide attendance without bypassing workflow.
AC-14	Reports	REVIEW/UNKNOWN are not ordinary attendance outcomes.
AC-15	Threshold	Low-attendance reporting uses institution-configured threshold.
AC-16	Audit	Relevant changes are recorded and audit history is read-only.

AC-17	Pagination	Large lists load incrementally with filtering before pagination and no missing/duplicate records.
AC-18	Platforms	Core workflows adapt to Web, Android and iOS.
AC-19	Frontend biometric boundary	Frontend does not generate embeddings, inspect vectors, store biometric templates or perform recognition.

22. Final Consistency Check

The SRS has been consolidated around the finalized requirement that faculty selects one or more assigned classes. This rule is applied consistently to the introduction, scope, workflow, functional requirements, recognition candidate scope, database requirements, twin handling, reports, use cases and acceptance criteria.

Principle	Covered In
One or multiple assigned classes	Sections 1, 4, 5, 15, 17, 21
Selected-class-only candidates	Sections 1, 4, 5, 7, 15, 21
Twin review	Sections 4, 6, 7, 17, 21
AI/final separation	Sections 1, 6, 16, 21
Faculty final authority	Sections 1, 5, 16
Post-finalization editing	Sections 4, 16, 17, 21
Admin workflow restriction	Sections 11, 12, 16, 21
Auditability	Sections 4, 13, 21
Deferred date-range/export	Sections 10, 20

23. Final Summary

- Single classroom photo → automatic face detection/recognition assistance.
- One or multiple assigned classes → class-scoped candidate pool.
- Unselected classes cannot contribute candidates.
- Existing normalized database remains the foundation.
- Frontend only provides face-enrollment UI/status; biometric processing belongs to backend.
- AI is advisory; faculty is final authority.
- PRESENT, ABSENT, UNKNOWN and REVIEW retain defined meanings.
- Twin ambiguity remains REVIEW until faculty decision.
- Faculty can edit before and after finalization with appropriate controls.
- AI result remains preserved while final result changes when faculty edits.
- Admin has institution-level oversight but cannot bypass the established attendance workflow.
- Student, faculty, class, history, reports, audit and settings are covered.
- Pagination is required for large collections.
- Web, Android and iOS are supported.
- Report date-range UI and export remain deferred/planned.